

A2 Warm Up Compound Interest & Graphing Exp and Log Equations

Remember not to round until the very end

1. Bill invests \$2977 in a retirement account with a fixed annual interest rate of 6% compounded quarterly. What will the account balance be after 13 years, rounded to the nearest hundredths.

$n=4$

$t=13$

$$A = 2977 \left(1 + \left(\frac{.06}{4} \right) \right)^{(4 \times 13)} = \$6,456.74$$

2. Talia invests \$2476 in a savings account with a fixed annual interest rate of 4% compounded monthly. What will the account balance be after 6 years, rounded to the nearest cent?

$n=12$

$$A = 2476 \left(1 + \left(\frac{.04}{12} \right) \right)^{(12 \times 6)} = \$3,146.36$$

3. Graph $y = 3(2)^x - 4$ remember: asymptote, 2 special points (exp. = 0, exp. = 1), and shape

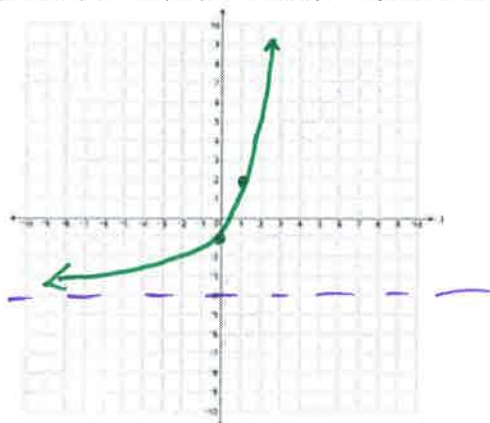
$b > 1$ and exponential growth!

BASIC SHAPE

asymptote @ $y = -4$

$$\begin{aligned} \text{exp} = 0 \\ x = 0 \\ y &= 3(2)^0 - 4 \\ y &= 3 - 4 = -1 \\ (0, -1) \end{aligned}$$

$$\begin{aligned} \text{exp} = 1 \\ x = 1 \\ y &= 3(2)^1 - 4 \\ y &= 6 - 4 = 2 \\ (1, 2) \end{aligned}$$



4. Graph $y = \log_{10}(x + 3) - 2$ asymptote, 2 special points: () = base, () = 1, and shape

log graph! Basic shape: Asymptote $x = -3$

$$\begin{aligned} () = \text{base} \\ x + 3 &= 10 \\ x &= 7 \end{aligned}$$

$$\begin{aligned} y &= \log_{10}(7+3) - 2 \\ y &= 1 - 2 \\ y &= -1 \\ (7, -1) \end{aligned}$$

$$\begin{aligned} () &= 1 \\ x + 3 &= 1 \\ x &= -2 \end{aligned}$$

$$\begin{aligned} y &= \log_{10}(-2+3) - 2 \\ y &= \log_{10}(1) - 2 \\ y &= 0 - 2 = -2 \\ (-2, -2) \end{aligned}$$

